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Boost converter design for a 3 MW solar PV system

Sizes L, C, R and duty cycle for the boost stage.

```
clear; clc;
```

Design inputs

```
P    = 3e6;       % Rated power (W)
Vin  = 383.5;     % Converter input voltage (V)
Vout = 600;       % Required output voltage (V)
fs   = 10e3;      % Switching frequency (Hz)

ripple_I = 0.01;  % Inductor-current ripple (1% of input current)
ripple_V = 0.01;  % Output-voltage ripple (1% of output voltage)
```

Currents and ripple targets

```
Iin   = P/Vin;               % Input (inductor) current (A)
Iout  = P/Vout;              % Output current (A)
dIL   = ripple_I*Iin;        % Inductor-current ripple (A)
dVout = ripple_V*Vout;       % Output-voltage ripple (V)
```

Design equations

```
D = (Vout - Vin)/Vout;                  % Duty cycle
L = (Vin*(Vout - Vin))/(dIL*fs*Vout);   % Inductance (H)
C = (Iout*(1 - Vin/Vout))/(fs*dVout);   % Capacitance (F)
R = Vout/Iout;                          % Equivalent load resistance (ohm)
```

Results

```
fprintf('Input current   Iin  = %.1f A\n',  Iin);
fprintf('Output current  Iout = %.1f A\n',  Iout);
fprintf('Current ripple  dIL  = %.2f A\n',  dIL);
fprintf('Duty cycle      D    = %.4f (%.2f %%)\n', D, D*100);
fprintf('Inductance      L    = %.4e H (%.2f uH)\n', L, L*1e6);
fprintf('Capacitance     C    = %.4f F (%.2f mF)\n', C, C*1e3);
fprintf('Load resistance R    = %.4f ohm\n', R);
```

```
Input current   Iin  = 7822.7 A
Output current  Iout = 5000.0 A
Current ripple  dIL  = 78.23 A
Duty cycle      D    = 0.3608 (36.08 %)
```

```
Inductance      L    = 1.7690e-04 H (176.90 uH)
Capacitance     C    = 0.0301 F (30.07 mF)
Load resistance R    = 0.1200 ohm
```

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